Experiment-4: Setting Up a Local Area Network

3/8/26

1) Aim:

To design, set up and configure a Local Area Network (LAN) connecting four computers through a network switch assign static IP address to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

2) Equipment Required

S.NO	Equipment	Quantity	Specification
1.	Personal Computer (PC)	4	Ethernet (NIC) enabled.
2	Network Switch	1	8/16/24 port, unmanaged/managed
3.	Ethernet Cables (Cat5e/cat6)	4	Straight-through, RJ-45 connectors.

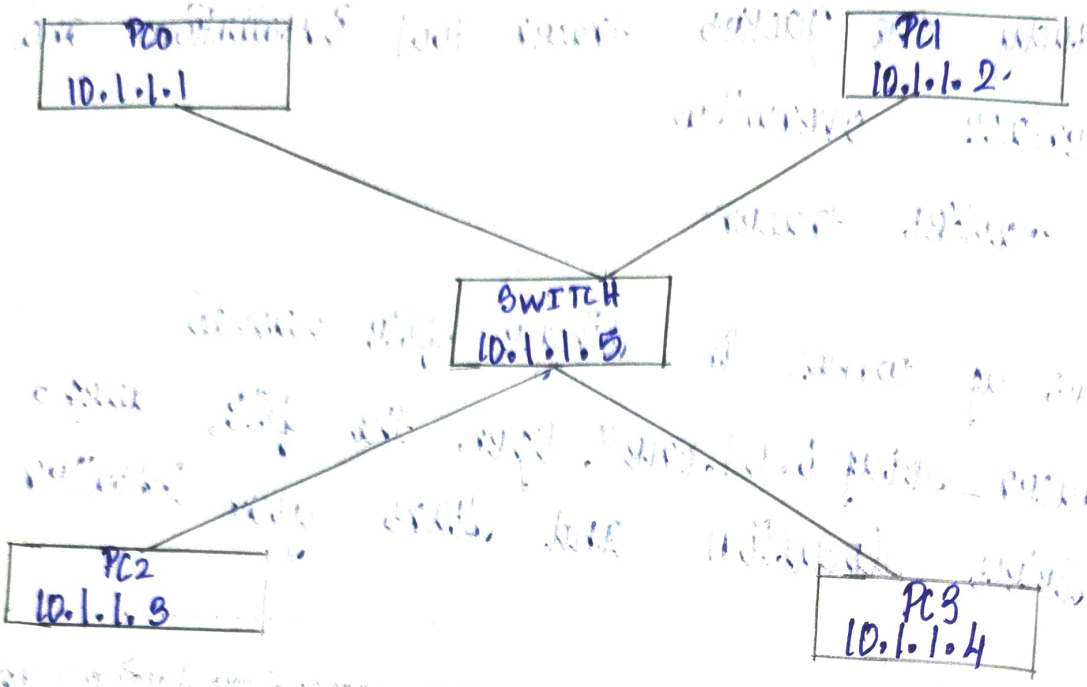
3) Theory:

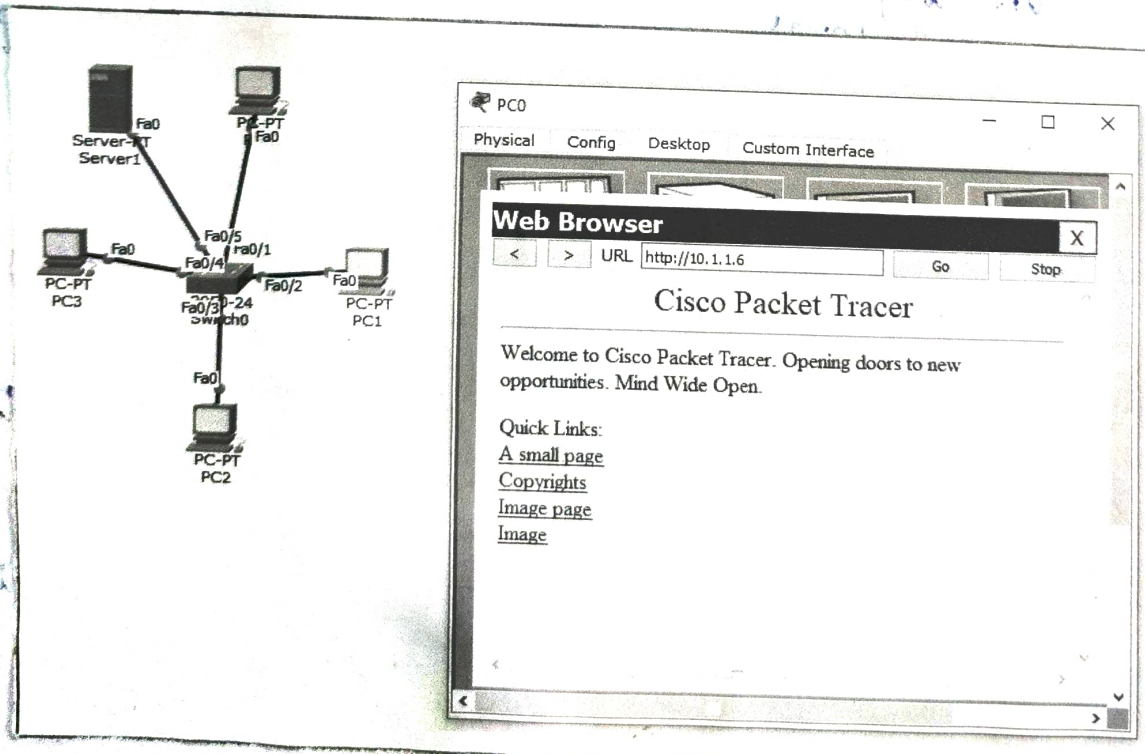
A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area such as a home, office, or laboratory, allowing them to share resources such as files, printers, and internet connections.

4) Configuration Table.

Device	IP address	Subnet Mask	Connected To.
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface.

STAR Topology LAN (4 PCs + 1 Switch)





Output

6. Implementation in cisco Packet Tracer

Since physical hardware was limited, the LAN was also implemented and verified using Cisco Packet Tracer simulation software, following the steps below

Step 1: Place the device.

1) Open Cisco Packet Tracer
2) From the device category panel, select End devices and drag 4 PCs onto the workspace.
3) Select switches and drag 1 switch.

Step 2: Cable the devices

4) Click Connections → select copper straight-Through cable
5) Click each PC, then click the switch and select an available FastEthernet port to complete each connection.
6) Repeat this process for all four PCs.

Step 3: Assign IP Address to PCs.

7) Click a PC → Desktop tab → IP configuration
8) Enter the static IP address and subnet mask as per the IP table (Section 5) for each PC.

Step 4: Configure the switch management IP (optional).

9) Click the switch → CLI Tab.
10) Enter these commands

```
#enable
#Configure Terminal
#Interface vlan 1
#ip address 10.1.1.2 255.0.0.0
#No shutdown
#exit
```

Step 5: Verify Connectivity Using Ping.

11) Click a PC → Desktop tab → Command Prompt.

12) Use the Ping command to test connectivity

13) Successful "Reply from" message confirm connectivity

Step 6:

14) Add a server device

15) Assign it an IP address

16) Open the Server's Services.

Step 7:

18) Switch to Simulation mode

19) Send a ping or web request between two devices and use the Play / Capture.

Result:

Hence the simulation to view the message in another node is successfully executed.